



Benthic diatom community response to the sudden rewetting of a coastal peatland

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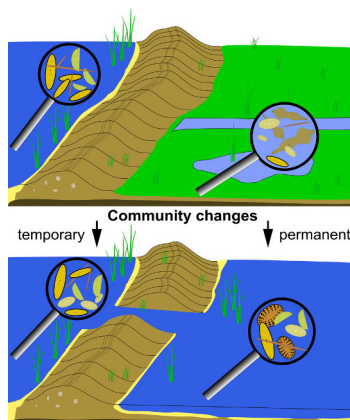
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HIGHLIGHTS

- Rewetting of a degraded coastal peatland releases nutrients into the adjacent lagoon.
- Targeted flooding event causes changes in diatom communities of peatland and lagoon.
- A taxonomic list of diatoms of a coastal peatland and Baltic Sea lagoon is presented.
- Benthic diatoms may be used as bio-indicators for dynamic coastal processes.

GRAPHICAL ABSTRACT



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ABSTRACT

The coastline of the shallow southern Baltic Sea is a highly dynamic system of spits, lagoonal systems and subtidal flats. On the land side, coastal peatlands are common. Intact peatlands provide important ecosystem services, however, many peatlands have been severely degraded due to human activities. Many coastal peatlands are also separated from the Baltic Sea by artificial dunes or dykes in order to facilitate drainage and coastal protection measures. Due to their potential to act as carbon sinks, as well as buffer zones against sea level rise and flooding events, there is an increasing interest in the restoration and rewetting of coastal peatlands. Microphytobenthic communities are important primary producers in coastal systems and peatlands and diatoms are often the dominant microphytobenthic group in temperate regions. In November 2019, the removal of the dyke that separated the Polder Drammendorf (a drained, oligohaline peatland), from the adjacent Baltic Sea lagoon Kubitzer Bodden represented a unique chance to study the sudden rewetting of a degraded coastal peatland with brackish water and its effect on the microphytobenthos. The aims of this study were to investigate the poorly

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known diatom community composition of coastal peatlands, to determine the effects of flooding events on diatom composition and biomass in coastal peatlands as well as the effects of nutrient and substance release from the peatlands on the diatom communities of adjacent coastal waters. We investigated and documented the impact of this sudden rewetting on the benthic diatom communities of the peatland and the adjacent lagoon over the course of one year, applying the latest taxonomy and measuring key environmental factors. The results deepen the understanding of diatom community composition, taxonomy and ecology and explore the influence of land-sea exchange processes on the benthic diatoms of the Baltic Sea coast.

1. Introduction

The Baltic Sea is a brackish shelf sea in Northern Europe, which is almost completely enclosed by the landmasses of the European mainland and the Scandinavian Peninsula. Due to this semi-enclosed character, the water exchange with the adjacent marine North Sea is poor. This in turn results in long water renewal times and a stable horizontal salinity gradient ranging from freshwater conditions in the Northern Gulf of Bothnia to mesohaline conditions in the western Baltic Sea (Busse and Snoeijs, 2002). Its specific hydrography causes the Baltic Sea to be vulnerable to increasing eutrophication, which started in the 1950s and has mainly been caused by the intensive agricultural land use in the southern drainage area (e.g. Cederwall and Elmgren, 1990). With tidal ranges of only 12–15 cm in the western part to mostly <10 cm in the remaining Baltic Sea, it can be characterised as a microtidal system (Snoeijs-Leijonmalm and Andrén, 2017). However, atmospheric pressure and wind conditions can cause distinct and swift water level changes as well as storm flood events (Lass and Magaard, 1996; Karsten et al., 2012). Therefore, the coastline of the shallow southern Baltic Sea is a highly dynamic system. Irregular water level changes, erosion and coastal currents lead to the dynamic transport, resuspension and deposit of sediments, shaping a constantly changing coastline of spits, lagoonal systems and subtidal flats.

Along the southern Baltic Sea coastline, coastal peatlands shape the transition zone of the terrestrial side to the Baltic Sea. Due to oxygen deficiency in peatlands (organic wetlands) plant production rates are higher than those of plant decomposition, resulting in the accumulation of partially decomposed organic matter called peat. Consequently, intact peatlands are important and massive carbon storages and sinks, representing approximately 30 % of the global carbon (Grützner, 2009). In contrast, drained peatlands emit the greenhouse gases CO₂, CH₄ and N₂O, thereby not only increasing their atmospheric concentrations but also contributing to global warming (Martikainen et al., 1993; Regina et al., 1996; Strack, 2008; Kaat and Joosten, 2009). Under natural conditions, the dynamic interface of marine and terrestrial ecosystems influences the water level and chemistry of coastal peatlands (Freund et al., 2004). However, many peatlands, including coastal fens, have been severely degraded due to peat cutting and drainage for agricultural use or urban expansion (Charman, 2002; Gaiser and Rühland, 2010). In addition, coastal peatlands are vulnerable to the effects of climate change especially in connection to rising sea level and sinking coasts (Jurasinski et al., 2018; Waller and Kirby, 2021). At present, many coastal peatlands are separated from the southern Baltic Sea by dunes or dykes in order to facilitate drainage and coastal protection measures. As coastal peatlands can provide valuable ecosystem services, such as acting as carbon storages and sinks, and mitigating against the effects of sea level rise and flooding events (Gaiser and Rühland, 2010; Schuerch et al., 2018), there is an increasing interest in the restoration and rewetting of degraded peatlands (Vasander et al., 2003; Waller and Kirby, 2021). In drained peatlands, due to the exposure to oxygen, the upper peat layer becomes mineralized and nutrient enriched (Smolders et al., 2006; Geurts et al., 2010). Therefore, rewetted peatlands may become, at least at the beginning, a source of nutrients, especially concerning NH₄⁺ and PO₄³⁻ (Lamers et al., 2002; Duhamel et al., 2017; Pönisch et al., 2023). It is also known that rewetted peatlands are a source of CH₄, a highly potent greenhouse gas (Günther et al., 2020). In

a coastal setting, however, marine water can be used for the rewetting, which might in turn reduce CH₄ production due to stimulated bacterial sulphate reduction (Pönisch et al., 2023, and references therein).

Rather than a separation line, shallow coasts represent marine-terrestrial gradients in which the interactions between shallow coastal waters and adjacent peatlands are among the potentially most important exchange processes along the southern Baltic Sea coastline (Jurasinski et al., 2018; Janssen et al., 2019). Connecting underground peat layers, terrestrial runoff, submarine groundwater discharge and weather-induced flooding events can facilitate the exchange of organic material, minerals, nutrients as well as (micro)organisms (Jurasinski et al., 2018; Pörlle et al., 2019). This in turn will modify the biogeochemical processes in both systems, the extent of which, however, is unknown (Janssen et al., 2019). The scale and frequencies of exchange processes along these marine-terrestrial gradients are also expected to increase, facilitated by a projected sea-level rise and an increase of storm surge events (Jurasinski et al., 2018; Pönisch et al., 2023, and references therein). Restoration and rewetting of coastal peatlands will equally modify or even increase such exchange depending on the chosen measures. These expected processes will interact with the biocenoses of both systems across the terrestrial shallow water interface, such as the diversity and function of the ecologically highly important microphytobenthos. However, nature and scale of these effects are so far poorly understood (Janssen et al., 2019).

The microphytobenthos is generally composed of diverse communities of various phototrophic bacteria and microalgae, specifically cyanobacteria, diatoms, chlorophytes, dinoflagellates and euglenids depending on the habitat environmental conditions (Colijn and De Jonge, 1984; Sundbäck et al., 2000; Launeau et al., 2018). Although often neglected (Hope et al., 2019), these communities can make up about 30 % of the total coastal primary production (Ask et al., 2016; Cahoon, 1999; Gerbersdorf et al., 2005; Glud et al., 2009; Karsten et al., 2021; Kuriyama et al., 2021; Schreiber and Pennock, 1995). As such, they are important constituents of the respective food webs, for example, providing a source of high quality carbon and essential fatty acids for higher trophic levels such as invertebrate grazers or deposit filter feeders (Hope et al., 2019). Microphytobenthic communities are also highly important with regard to the biostabilization and biogeochemistry of the sediments which they inhabit (Hussain and Khoja, 1993; Pinckney et al., 1995; Hope et al., 2019). With the excretion of sticky extrapolymeric substances (EPS) sediment particles are glued together, thereby modifying and stabilizing sediment surfaces against hydrodynamic forces (Krumbein et al., 1994). The microphytobenthic communities represent a biofilm and hence a control barrier for the vertical exchange of dissolved gases and nutrients at the sediment-water column interface (Kuriyama et al., 2021).

In temperate regions, diatoms are often the dominant microphytobenthic group (Cahoon, 1999; Hope et al., 2019; Kuriyama et al., 2021). They are highly diverse and important components of carbon, nitrogen, phosphorus and silicate cycles at the sediment-water column interface (Cahoon, 1999; Morel and Price, 2003; Wilhelm et al., 2006; Ask et al., 2016; Pörlle et al., 2021). Diatoms, which form and inhabit sediment biofilms, can be either highly motile and respond to environmental gradients by moving horizontally and vertically through the upper sediment layer (epipelic) or they are weakly to non-motile and living rather attached to sediment grains (epipsammic).

Despite a relatively rich literature on benthic diatom ecology and community composition of the Baltic Sea (e.g. Weckström and Juggins, 2006; Ulanova et al., 2009; Witak, 2013; Virta and Soininen, 2017) estuarine and lagoonal systems (e.g. Juggins, 1992; Underwood et al., 1998; Rovira et al., 2012) and salt marshes (e.g. Sullivan and Currin, 2002; Trobajo et al., 2004; Roe et al., 2009), communities of coastal peatlands and the effects of land-sea exchange processes on benthic diatom communities are largely unknown. Moreover, due to an ever-evolving taxonomy and new taxonomical methodologies (e.g. molecular taxonomy) ecological inferences in older studies may be affected by an incomplete and outdated taxonomy. For these reasons taxonomic assessments have to be updated continuously.

In November 2019, the removal of the dyke that separated the polder in Drammendorf (a drained, oligohaline peatland in Northern Germany), from the adjacent Baltic Sea lagoon Kubitzer Bodden represented a unique chance to study the controlled sudden rewetting of a degraded coastal peatland with brackish water and its effect on the microphytobenthic community composition (Janssen et al., 2019; Pönisch et al., 2023). Therefore, in this study we investigated, for the first time, the spatial and temporal impact of this sudden rewetting on the benthic diatom communities of the peatland and the adjacent lagoon just before and after the event. This allowed for a first glimpse into the desired and potentially unwanted effects of rewetting peatlands, besides the so far well understood increase in carbon storage capacity. We aimed to deepen the understanding of benthic diatom community composition and ecology along marine-terrestrial gradients, and to explore the influence of land-sea exchange processes on the microphytobenthos of the southern Baltic Sea coast.

2. Methods

2.1. Study site

The studied coastal peatland ($54^{\circ}22'19.1''\text{N}$, $13^{\circ}14'30.6''\text{E}$), named Polder Drammendorf, is situated at the north eastern German Baltic Sea coast, on the western part of the island of Rügen (state of Mecklenburg-Western Pomerania), bordering the lagoon Kubitzer Bodden (Fig. 1). These localities are hereafter referred to as peatland and lagoon, respectively. With an average depth of 1.5 m and a salinity of 8.5 (Pönisch et al., 2023), the lagoon is a shallow, mesohaline aquatic environment. It is part of a larger lagoonal system and immediately connected to other coastal waters like Strelasund and Schaproder Bodden. Within the period of investigation (2019–2020) the peatland was transformed from a highly degraded, oligohaline coastal peatland to a mesohaline wetland, which is now directly connected to the lagoon and hence partially influenced by marine sediments. In the 1960s, the peatland had been artificially drained and cut off from the lagoon via the construction of a dyke (Pönisch et al., 2023). Subsequently, different parts of the drained peatland were used for various agricultural purposes like pasture and farmland. As a consequence, the top 50–70 cm of the soil consisted of highly degraded peat (Brisch, 2015), which was not removed prior to the rewetting. Underneath the degraded layer lies a well-preserved peat layer with a thickness of around 1 m and up to 2.2 m (Pönisch et al., 2023). In November 2019, the peatland was partially flooded with lagoonal water by the targeted removal of a 20 m section of the dyke. The height of the remaining dyke was reduced to the surface level. In total, the rewetted area covers ca. 0.8 km^2 and includes a

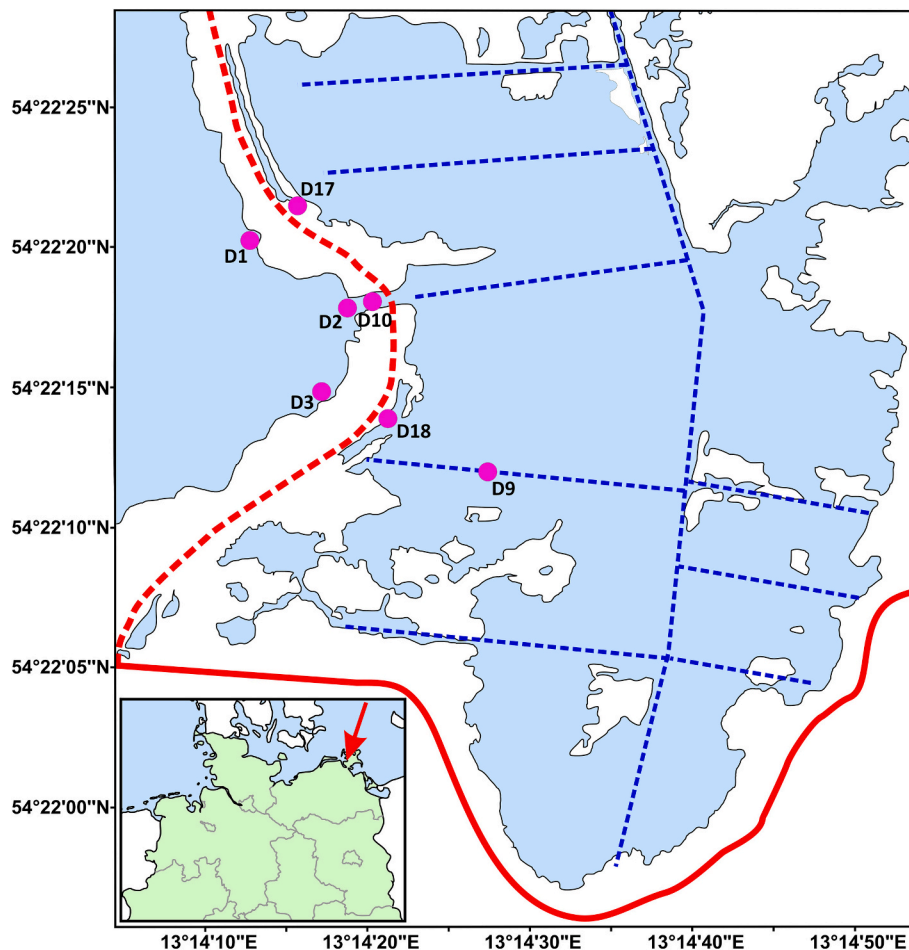


Fig. 1. Map of the study area after the flooding. Magenta dots indicate the sampling sites with names. Dark blue lines show drainage ditches. Red lines indicate dykes. Dashed lines represent pre-flooding dyke structures. The red arrow indicates the position of the study area in Northern Germany.

permanently water-covered area of ca. 0.5 km² with an average water depth of around 0.5 m (Pönisch et al., 2023).

2.2. Sampling of microphytobenthic communities

All sampling locations are indicated in Fig. 1. Sampling locations D1–D3 are situated at the edge of the lagoon and D9, D17 and D18 are located behind the dyke, inside the peatland. D10 is located in the canal which was created during the rewetting process and which now directly connects the lagoon and the rewetted peatland. D2 lies directly in front of the breakthrough. Samples were taken in October 2019, few weeks before the opening of the dyke (November 2019), approximately two months after the opening in January 2020 and again one year after the first sampling in October 2020. At each of the sampling times, three replicate samples were taken at each location. D1 and D3 were sampled at all three occasions. D2 could not be sampled in October 2019 due to preparations of the dyke breakthrough and the sampling of D9 was not possible in January 2020 due to personal risk. At the sampling locations D10, D17 and D18 no sediments existed prior to the opening (only pasture) and consequently these locations were only sampled after the dyke removal. All samples were taken in Plexiglas® core liners (Ø50 mm internal diameter) and always the top 10 mm of each core were used for the following analyses.

2.3. Environmental variables

The upper 10 mm sediment layer of each sample was carefully removed using a spatula, afterwards manually homogenised and divided into subsamples. Subsamples were either fixed with Lugol and stored for later preparation of permanent slides, or frozen for chlorophyll *a* analyses as an indicator for biomass using a fixed volume of 1.13 cm³. Chlorophyll *a* was extracted using 3 ml of 96 % (v/v) ethanol and incubated at 72 °C for 30 min. Extracts were centrifuged (Heraeus Megafuge 1.0R, Hanau, Germany) for 10 min at 4660 × *g* and supernatants measured at an extinction of 750, 696, 665, 649 and 632 nm with a spectrophotometer (Shimadzu UV-2401 PC, Kyoto, Japan). Chlorophyll *a* concentrations were calculated according to Ritchie (2008).

Salinity, pH, oxygen and dissolved inorganic nutrients (PO₄^{3−}, NH₄⁺, NO₃[−]; NO₂[−]) were measured from the surface water at the respective study sites. Salinity and oxygen measurements were measured in situ using a pre-calibrated multimeter (HQ 40d Multi, Hach Lange, Düsseldorf, Germany). Water samples were taken to analyse the other environmental variables *ex situ*. pH was measured using a three-way calibrated pH-meter (Sevenmulti, Mettler Toledo, Columbus, USA). For dissolved inorganic nutrient analysis, water samples were filtered through Whatman GF/6 glass fibre filter (Ø 25 mm) and frozen at −20 °C until further analysis. Nutrients were later analysed photospectrometrically following the approach described in Prelle et al. (2019).

2.4. Diatom sample preparation, identification and counting

The sediment subsamples mentioned under 2.3. were carefully cleaned for the preparation of permanent slides. To remove salts, each sediment sample was repeatedly cleaned by adding ultrapure water and subsequent gentle centrifugation at 1410 × *g* for 6 min. After discarding the liquid for the last time, the remaining sediment was treated with 30 % hydrogen peroxide for approximately two weeks at a room temperature to clean the diatom valves. The peroxide residues were again removed by repeated rinsing with ultrapure water and centrifugation as described above. Cleaned valve suspensions were transferred to cover slips using micro-pipettes and mounted on permanent slides using Naphrax®. The slides were afterwards examined using a Zeiss Axio Scope.A1 microscope with a Zeiss Plan-Apochromat 100×/1.4 DIC oil immersion objective. On average, 348 valves per sample were counted along random transects on each slide to calculate the relative abundance of diatom taxa present. Taxa with an abundance of >5 % in at least one

sample were classified as abundant.

For the morphological diatom identification, the following literature was mainly used: Krammer and Lange-Bertalot (1986, 1988, 1991), Snoeijs (1993), Snoeijs and Vilbaste (1994), Snoeijs and Potapova (1995), Snoeijs and Kasperovičienė (1996), Snoeijs and Balashova (1998), Witkowski (1994), Witkowski et al. (2000) and Lange-Bertalot et al. (2017). The current nomenclature according to Guiry and Guiry (2024) was used. In case of uncertain identification or unknown taxa, the following taxonomic terminology was used: *confer* (cf.) for an identification with uncertainty; *affinis* (aff.) for taxa that we believe resemble the given name but are in fact a different, unknown, taxon and *aggregate* (agg.) for a group of similar taxa that could or were not differentiated (e.g. cryptic species).

2.5. Data analysis

Correlations between the different environmental parameters were evaluated by calculating the Spearman's rank correlation coefficients using R (ggpubr and dplyr, Wickham et al., 2023, Kassambara, 2023). In order to compare the sample and community compositions a detrended correspondence analysis (DCA), using relative species abundances, was performed in R (vegan, Oksanen et al., 2022). Down-weighting was applied to reduce the influence of rare species. To analyse the relationships between species and communities and environmental parameters, canonical correspondence analyses (CCA) were performed using R (vegan, Oksanen et al., 2022). A global CCA for the complete dataset (all samples and parameters) was performed using forward selection to avoid overparameterization. The variance inflation factor (VIF) was tested to be below 10 in order to minimize multicollinearity. For the CCA of the species, only species with an abundance of >5 % in at least one sample are shown by name. The respective abbreviations are listed in Table 1. To investigate the question of how the dyke opening and subsequent flooding influenced the lagoon and peatland communities, we focused on changes in nutrients and salinity as these are known to be important for the community compositions (e.g. Snoeijs, 1994, 1995; Weckström and Juggins, 2006; Ulanova et al., 2009) and at the same time directly influenced by the flooding (Pönisch et al., 2023). Therefore, two separate CCAs were performed, each focusing on nutrients (PO₄^{3−}, NH₄⁺, NO₃[−]) and salinity in the lagoon and peatland, respectively. Forward selection and a VIF below 10 were equally applied in this case. A Venn diagram for the peatland and lagoon communities before and after the flooding was created using R (ggvenn, Yan, 2023). Concerning the chlorophyll *a* concentrations, a Kolmogorov-Smirnov and a Shapiro-Wilks test of normality were performed as the contents of each defined communities can not all be assumed normally distributed ($\alpha = 0.05$). Therefore, a Kruskal–Wallis test was performed to test for differences between the different times/communities. Information about species traits was taken from Vos and de Wolf (1993), Vilbaste et al. (2000), Trobajo et al. (2004), Weckström and Juggins (2006), Ulanova et al. (2009), De Sève et al. (2010), Majewska et al. (2013), Virta and Soininen (2017) and Spaulding et al. (2021). In addition to species richness, the Gini-Simpson diversity index was calculated for peatland and lagoon at each of the three sampling times as

$$D = 1 - \sum_{i=1}^R \frac{n_i(n_i - 1)}{N(N - 1)}$$

where *R* is the species richness, *N* the total number of counted valves and *n_i* the number of valves of species *i*.

3. Results

3.1. Environmental parameters

Rewetting had a strong effect on the measured physico-chemical parameters (Table 2). While PO₄^{3−} and NO₂[−] concentrations were quite

Table 1

List of taxa with over 5 % relative abundance in at least one sample. Average relative abundances in % are given for lagoon and peatland at the three different sampling times. Sampling location D2 is listed separately in October 2020, as the location and community composition is intermediate. LF = Life form; B = benthic, P = planktonic, EPE = epipelagic, EPS = epipsammic, EPY = epiphytic. Motility; M = motile, N = non-motile, W = weakly motile.

Aerophil	LF	Motility	Taxon	Code	October 2019		January 2020		October 2020		
					Lagoon	Peatland	Lagoon	Peatland	Lagoon	D2	Peatland
–	EPY	W	<i>Berkeleya rutilans</i> (Trentepohl ex Roth) Grunow	B.rut	1.7		7.1	0.6			0.2
+	B	M	<i>Chamaepinnularia krookiformis</i> (Krammer)	C.kro	0.8	0.1	1.7	5.9	0.2		0.3
–	EPY	W	Lange-Bertalot & Krammer <i>Cocconeis lineata</i> Ehrenberg	C.pla. lin	5.6	0.2	1.7	1.9	1.4	0.7	0.4
–	EPY	W	<i>Cocconeis</i> sp. B	C.sp.B	0.8		0.1		2.5	0.6	0.2
–	P	N	<i>Cyclotella atomus</i> Hustedt	C.ato		0.1			2.3	7.0	3.8
–	P	N	<i>Cyclotella choctawhatcheeana</i> Prasad	C.cho	0.5				1.8	5.3	2.8
–	EPY	M	<i>Denticula subtilis</i> Grunow	D.sub	1.4		6.9	1.8	0.7	0.5	0.1
+	EPE	M	<i>Diploneis interrupta</i> (Kützing) Cleve	D.int	1.8	2.5	0.5	5.4	0.5		0.4
–	EPE	M	<i>Fallacia clepsidroides</i> Witkowski	F.cle	0.1				1.7		0.2
–	EPS	N	<i>Fragilaria amicornum</i> Witkowski & Lange-Bertalot	F.ami	1		0.5	0.1	1.6	0.3	0.3
–	EPS	N	<i>Fragilaria cassubica</i> Witkowski & Lange-Bertalot	F.cas	1.5		0.2	0.1	1.2	0.7	1.0
–	EPS	N	<i>Fragilaria gedanensis</i> Witkowski	F.ged	4.8	0.1	0.4	1.4	3.0	1.4	0.6
–	B	M	<i>Halamphora acutiuscula</i> (Kützing) Levkov	H.acu	0.6		0.3		0.3		0.6
–	EPY	W	<i>Karayevia</i> aff. <i>Amoena</i> (Hustedt) Bukhtiyarova	K. amoe			0.1		2.0	0.2	
+	EPE	M	<i>Luticola acidoclinata</i> Lange-Bertalot	L.aci				2.7			0.1
–	EPS	N	<i>Martyana atomus</i> (Hustedt) Snoeijjs	M.ato	2.4	0.2	1.1	2.1	1.2	1.7	3.8
+	EPE	W	<i>Mayamaea</i> cf. <i>permitis</i> (Hustedt) Bruder & Medlin	M.per		5.0					
–	EPS	N	<i>Nanofrustulum krumbeinii</i> (Witkowski, Witak & Stachura) Morales	N. krum	3.3	0.8	1.9	1.0	6.0	9.0	13.9
–	B	N	<i>Nanofrustulum trainorii</i> (Morales) Morales	N.trai	1.2	0.4	1.8		4.8	2.7	1.2
–	EPE	M	<i>Navicula gregaria</i> Donkin	N.gre	8.3	0.2	3.1	1.1	1.5	1.5	2.2
–	EPE	M	<i>Navicula normaloides</i> Cholnoky	N. norm	1.9		0.5		0.3		0.1
–	EPE	M	<i>Navicula paul-schulzii</i> Witkowski & Lange-Bertalot	N.p-s	4.1		2.8	0.5	3.1	1.5	0.8
–	EPE	M	<i>Navicula perminuta</i> Grunow	Na.per	6.0		5.2	1.2	2.4	13.7	19.0
–	EPE	M	<i>Navicula salinarum</i> Grunow agg.	N.sala	0.2		1	0.6		0.3	1.5
–	EPE	M	<i>Navicula salinicola</i> Hustedt	N.salc	3.9		0.7			0.4	0.4
–	EPE	M	<i>Nitzschia inconspicua</i> Grunow	N.inc	0.8		0.9		2.1	2.1	1.2
–	EPE	M	<i>Nitzschia</i> aff. <i>Microcephala</i> Grunow	N.mic	1.6					0.1	2.2
–	EPE	M	<i>Nitzschia nana</i> Grunow	N. nana	0.3	7.1	2.8		0.2	0.1	0.1
–	EPE	M	<i>Nitzschia palea</i> (Kützing) Smith	N.pal		8.6					
–	EPE	M	<i>Nitzschia</i> aff. <i>Perindistincta</i> Cholnoky	Ni.per	0.1		2.4				
–	EPE	M	<i>Nitzschia</i> cf. <i>pusilla</i> Grunow	N.pus	1.2	39.9	0.2	0.3	0.5	0.6	8.3
–	EPE	M	<i>Nitzschia</i> sp. A	N.A	0.4		0.8	0.1	0.8		1.0
–	EPS	N	<i>Opephora</i> cf. <i>mutabilis</i> Sabbe & Wyverman	O.mut	3.7		2.6	1.9	6.9	2.9	0.5
+	EPE	M	<i>Pinnularia borealis</i> Ehrenberg agg.	P.bor	0.1	0.1	1.3	2.1			0.1
–	B	M	<i>Pinnularia streckeriae</i> Krammer	P.stre		4.0					
–	EPS	W	<i>Planothidium engelbrechtii</i> (Cholnoky) Round & Bukhtiyarova	P.eng		5.5		0.3	0.3	0.4	0.3
–	EPS	W	<i>Planothidium hauckianum</i> (Grunow) Bukhtiyarova	P.hau	1.4	0.2	0.6	0.3	3.3	0.6	0.4
–	EPY	W	<i>Planothidium minutissimum</i> (Krasske) Lange-Bertalot	P.min	1.8			0.2	2.4	0.2	0.1
–	EPS	W	<i>Planothidium septentrionale</i> (Østrup) Round & Bukhtiyarova	P.sept	0.6				2.8	0.3	0.2
–	EPS	W	<i>Planothidium</i> sp. “C”	P.sp.C	0.1		0.2		3.1		
–	EPS	N	<i>Pseudostaurisira bronkei</i> (Witkowski, Lange-Bertalot & Metzeltin) Wetzel & Morales	P.bro	0.5	0.6	1.4	0.6	1.7	0.2	0.2
?	EPS	N	<i>Nanofrustulum</i> cf. <i>lucectorii</i> Morales, Novais, Garcia, Maidana & Morais	N.luc	4.7	5.4	8.4	17.8	7.4	1.9	1.2
–	EPS	N	<i>Staurisirella guenter-grassii</i> (Witkowski & Lange-Bertalot) Morales, Wetzel & Ector	S.g-g	6.1		7.1	5.6	9.5	7.5	5.0
–	B	N	<i>Staurisirella pinnata</i> (Ehrenberg) Williams & Round	S.pin	0.6	1.0	1.8	2.3	1.2	1.8	1.3
–	EPE	M	<i>Surirella brebissonii</i> Krammer & Lange-Bertalot	S.bre			0.8	3.5			0.1
–	EPY	N	<i>Tabularia massenana</i> Heudre, Wetzel, Van de Vijver, Moreau & Ector	T.mas	0.3	0.5	2.4	0.1	0.2		0.3
–	P	N	<i>Thalassiosira pseudonana</i> Hasle & Heimdal	T.pseu					1.4	25.8	11.9

low in the peatland before rewetting, both nutrients strongly increased (10 to 57 times PO_4^{3-} , 4 to 9 times NO_2^-) after one year flooding. An opposite trend could be observed for NH_4^+ and NO_3^- , which exhibited high contents in the peatland before rewetting and lower ones after 12 months (Table 2). The lagoon stations showed before the dyke opening relatively low concentrations of all four nutrients, but enriched values after the opening and rewetting. Temperature and oxygen were not strongly affected by flooding, but pH and salinity. The former peatland changed from slightly acidic (pH 6.5) to more alkaline (pH 7.7)

conditions, and salinity strongly increased from 1.2 to 7.7–10.9 (Table 2). In many cases, the measured environmental variables were moderately or even strongly correlated (Table 3), which was especially true between NO_3^- , temperature and O_2 .

3.2. Chlorophyll a measurements

Chlorophyll a concentrations were measured as a proxy for photo-trophic biomass. The results for peatland and lagoon at the different

Table 2

Environmental variables in the peatland and lagoon at the three sampling dates.

	October 2019		January 2020		October 2020	
	Peatland	Lagoon	Peatland	Lagoon	Peatland	Lagoon
PO ₄ ³⁻ (μM)	0.01	0–0.18	0–0.04	0	0.1–0.57	0.19–0.26
NO ₂ ⁻ (μM)	0.3	0.26	0.63–1.77	0.54–0.83	1.18–2.80	2.84–4.54
NH ₄ ⁺ (μM)	245.6	1.2–1.4	3.2–52.0	2.2–8.0	42.1–183.8	21.5–35.2
NO ₃ ⁻ (μM)	21.9	0.18–0.19	29.3–57.7	17.6–30.4	0.8–5.8	6.8–10.1
Temperature (°C)	12.9	14.2–15.1	3.8–4.2	4.1–4.4	14.5–15.8	14.5–16.1
pH	6.5	7.4–7.6	7.9–8.0	8.1–8.2	7.6–7.8	7.5–7.9
Salinity (PSS-78)	1.2	8.4	7.1–7.3	7.4–7.5	7.7–10.9	7.5–8.0
O ₂ (mg/L)	13.5	8.8–9.8	13.1–13.4	12.4–12.7	9.2–10.0	8.5–11.1

Table 3

Spearman correlations coefficients between the measured environmental variables. *moderate correlation, **high correlation, ***very high correlation (Prion and Haerling, 2014).

	PO ₄ ³⁻	NO ₂ ⁻	NH ₄ ⁺	NO ₃ ⁻	Temperature	pH	Salinity	O ₂
PO ₄ ³⁻	1							
NO ₂ ⁻	0.35	1						
NH ₄ ⁺	0.35	-0.14	1					
NO ₃ ⁻	-0.51*	-0.14	-0.0004	1				
Temperature	0.61**	0.40*	0.28	-0.81***	1			
pH	0.05	0.11	-0.65**	0.23	-0.50*	1		
Salinity	0.56*	0.20	-0.45*	-0.34	0.16	0.67**	1	
O ₂	-0.50*	-0.44*	0.21	0.79**	-0.76**	0.11	-0.52*	1

sampling times are given in Fig. 2. The overall measured chlorophyll *a* concentrations varied strongly between 0.9 and 200 mg m⁻². Average values within the different communities ranged from 31.2 mg m⁻² (peatland, January 2020) to 107.0 mg m⁻² (peatland, October 2020). At each of the three sampling times the chlorophyll *a* contents of lagoon

and peatland were similar and not significantly different ($p > 0.05$). The values between October 2019 and October 2020 were also comparable and did not differ significantly ($p > 0.05$). However, a significant ($p < 0.05$) decrease of chlorophyll *a* in the peatland and the lagoon was observed in January 2020, two months after the flooding.

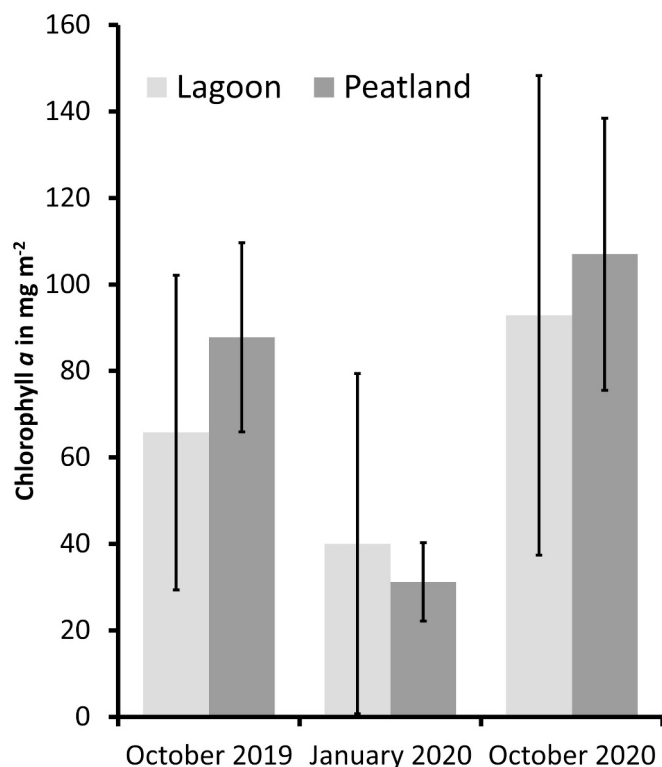


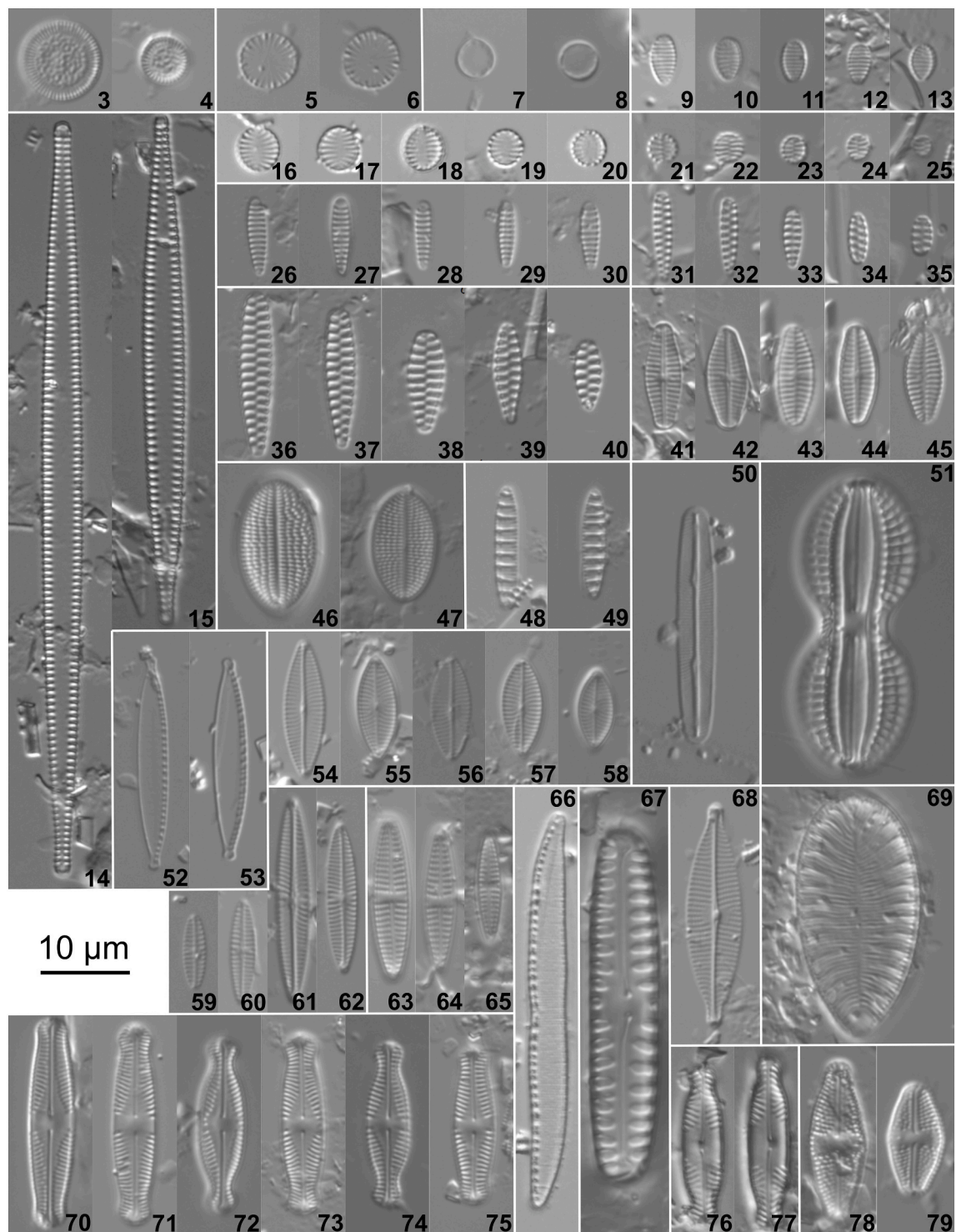
Fig. 2. Chlorophyll *a* concentrations in mg m⁻² for the lagoon and peatland areas over time. Error bars represent one standard deviation. Only the means from January 2020 were statistically significantly different from the values of October 2019 and 2020. At each sampling time peatland and lagoon were not significantly different.

3.3. Community compositions

Of the total of 48 samples, 39 samples could be successfully analysed. The remaining nine samples did not contain a sufficient number of diatoms and were therefore not included in the analyses. This mainly concerns samples taken shortly after the dyke opening in January 2020, and in particular all samples directly in front of the breakthrough (D2), two samples within the peatland (D18) as well as the samples taken at the newly created canal (D10) which now directly connects the lagoon and the peatland. In addition, one sample at D3 in October 2020 also did not contain a sufficient number of diatom valves.

In total, 245 different diatom species were found (Table S1, supplement). Among those, in 47 species the relative abundance was over 5 % in at least one sample. The taxonomic names and relative abundances of these regularly occurring taxa are given in Table 1 and micrographs of abundant and relevant taxa are presented in Figs. 3–79. Morphological species identification proved to be challenging in some cases and micrographs of abundant and uncertain identification are given in Fig. S1–S94 (supplement). Eight out of the 47 abundant taxa (17 %) could not be ambiguously identified, because of missing taxonomic information.

The peatland samples taken before the opening of the dyke (October 2019) were dominated by the oligohalobous-indifferent, epipellic *Nitzschia* taxa *Nitzschia* cf. *pusilla*, *N. palea* var. *palea* and *N. nana*. In addition, some halophilous to mesohalobous species, namely *Planorhynchus engelbrechtii*, *Diploneis interrupta* and *Ctenophora pulchella*, could be found. *Nanofrustulum* cf. *lucectorii* was another abundant taxon. The abundance of these taxa may therefore indicate some level of desiccation over the course of the year. In addition, *Pinnularia streckeriae*, a species reported from inland brackish waters, was relatively abundant and specific to the peatland before the flooding. At the same time, the lagoon was dominated by halophilous and mesohalobous, epipellic *Navicula* species (*Navicula gregaria*, *N. perminuta*, *N. paul-schulzii*, *N. salinicola*) as well as araphid, often epipsammic diatoms such as *Staurosirella guenter-*



Figs. 3–79. Diatom taxa with a relative abundance of at least 5 % in at least one sample. Figs 3–4: *Cyclotella choctawhatcheeana*, Figs 5–6: *C. atomus*, Figs 7–8: *Thalassiosira pseudonana*, Figs 9–13: *Martyana atomus*, Figs 14–15: *Tabularia massenana*, Figs 16–20: *Nanofrustulum trainorii*, Figs 21–25: *N. krumbeinii*, Figs 26–30: *Fragilaria gedanensis*, Figs 31–35: *Staurosirella guenter-grassii*, Figs 36–40: *Opephora* cf. *mutabilis*, Figs 41–45: *Planothidium engelbrechti*, Figs 46–47: *Cocconeis placentula* agg., Figs 48–49: *Denticula subtilis*, Fig. 50: *Berkeleya rutilans*, Fig. 51: *Diploneis interrupta*, Figs 52–53: *Nitzschia palea*, Figs 54–58: *Navicula paul-schulzii*, Figs 59–62: *N. salinicola*, Figs 63–65: *N. perminuta*, Fig. 66: *Nitzschia nana*, Fig. 67: *Pinnularia borealis* agg., Fig. 68: *Navicula gregaria*, Fig. 69: *Surirella brebissonii*, Figs 70–75: *Chamaepinnularia krookiformis*, Figs 76–77: *Pinnularia streckeriae*, Figs 78–79: *Luticola acidoclinata*.

grassii, *Fragilaria gedanensis*, *Nanofrustulum* cf. *lucectorii*, *Opephora* cf. *mutabilis*, *Nanofrustulum krumbeinii* and *Martyana atomus*. Additionally, the oligohalobous-indifferent taxon *Cocconeis placentula* var. *lineata* was also abundant.

In January 2020, approximately six weeks after the opening of the

dyke, *N. cf. lucectorii* became the most abundant taxon in the peatland area. It was accompanied by mesohalobous taxa such as *Chamaepinnularia krookiformis*, *S. guenter-grassii*, *D. interrupta* and *M. atomus* as well as the oligohalobous-indifferent taxa *Surirella brebissonii*, *Luticola acidoclinata*, *Staurosirella pinnata*, *Pinnularia borealis* agg. and *Hantzschia*

amphioxys. A number of these taxa are areophilic (*C. krookiformis*, *L. acidoclinata*, *P. borealis* agg. and *H. amphioxys*) indicating some level of desiccation. In parallel, an increase in the abundance of crysophyte cysts was observed. The diatom community of the lagoon at that time was still dominated by mesohalobous species. *N. cf. luectorii* and *S. guenter-grassii* were most abundant while *Navicula perminuta*, *N. gregaria*, *N. paul-schulzii*, *O. cf. mutabilis* were still frequent, *Berkeleya rutilans*, *Denticula subtilis*, *N. nana*, *Tabularia massenana* and *Nitzschia* aff. *Perindisticta* became abundant in this phase. Notably, oligohalobous-indifferent *N. nana* was formerly abundant in the peatland before the flooding.

In October 2020, the benthic community of the peatland was dominated by meso- to euhalobous taxa such as *Navicula perminuta*, *Nanofrustulum krumbeinii*, *S. guenter-grassii*, *M. atomus* and *Navicula gregaria*. Moreover, the return of *Nitzschia* cf. *pusilla* and the occurrence of *N. aff. microcephala* could be observed. In addition to these benthic species, planktonic centric taxa became abundant during this phase. These were dominated by euryhaline *Thalassiosira pseudonana*, oligohalobous-halophilous *Cyclotella atomus* and mesohalobous *C. choctawhatcheeana*. The high abundance of these planktonic taxa (13–39 %) seems to be limited to the peatland area and the immediate connection to the lagoon (D2), while relative abundances further into the lagoon (D1, D3) were relatively low (2–8 %). The benthic community of the lagoon at that time was dominated by small fragilaroid taxa such as *S. guenter-grassii*, *Nanofrustulum krumbeinii*, *O. cf. mutabilis*, *N. cf. luectorii*, *N. trainorii* and *F. gedanensis*. Concerning motile, epipellic species, *Navicula perminuta* and *N. paul-schulzii* were most abundant.

3.4. Detrended correspondence analysis

In the DCA, samples from the same sampling locations always clustered together and the overall spatial and temporal structure of the samplings is well captured (Fig. 80). At each sampling time, peatland and lagoon samples showed a high degree of similarity within the two areas, justifying the use of the terms peatland and lagoon community, respectively. Between the different sampling times, however, these communities can vary distinctly, exhibiting clear temporal phases. The peatland samples before the rewetting (October 2019) were distinct and most different from all other samples, including the lagoon samples at the same time. In January 2020, two months after the intentional

flooding of the peatland, both communities changed from their original and quite different compositions to an almost identical brackish community. Another nine months later, in October 2020, lagoon and peatland community compositions changed again. The community of the lagoon became again similar to its pre-flooding composition, while the peatland community changed another time to develop to a likewise, but still different lagoon community at that time. The sampling point D10 in October 2020 was colonized by benthic diatoms and belongs to the peatland community. D2, which lies directly in front of the opening, showed an intermediate composition, distinctly influenced by the peatland.

3.5. Canonical correspondence analysis

The relationship between communities/species and the environmental variables was analysed with the help of CCAs. The spatial and temporal changes of the measured environmental variables within the different communities are captured in Table 2. In order to minimize multicollinearity (VIFs), the parameter pH had to be removed from the global CCA. All further parameters were accepted by the forward selection process. The remaining parameters explained 40.9 % of the total variation (inertia) of the dataset. Overall, the global CCA for the sites (Fig. 81) showed a pattern consistent with the DCA, separating the same main phases/clusters. Axis 1 is related to NH_4^+ and salinity, mainly separating the pre-flooding peatland samples (October 2019), which exhibited a strong relationship with high NH_4^+ and low salinity. Axis 2 mainly separated the three sampling dates and is related to temperature, PO_4^{3-} and NO_3^- . The samples from January 2020 were related to low temperatures and high NO_3^- concentrations and the samples from October 2020 could be linked to higher temperatures, PO_4^{3-} and NO_2^- as well as lower O_2 and NO_3^- concentrations.

The global CCA for the species (Fig. 82) equally showed a distribution of taxa mainly along Axis 1 and 2. It is, however, a more continuous distribution compared to the sites. Concerning the abundant species among the pre-flooding peatland community, *Pinnularia streckeriae*, *M. cf. permits* and *Nitzschia palea* exhibited optima most different from the brackish communities, while *N. cf. pusilla*, *N. nana* and *Planothidium engelbrechtii* appeared to be more intermediate. *B. rutilans*, *N. aff. perindisticta*, *D. subtilis*, *S. brebissonii* and the aerophilic taxa *Pinnularia borealis* agg., *C. krookiformis* and *L. acidoclinata* showed optima associated with the conditions in January 2020. Among the dominant species in the peatland in October 2020, the planktonic species *C. atomus*, *C. choctawhatcheeana* and *T. pseudonana* exhibited similar optima and seemed to respond to elevated PO_4^{3-} concentrations. Among benthic diatoms, *N. aff. microcephala* and *Nanofrustulum krumbeinii* showed a preference for the conditions at that time. Most of the remaining abundant taxa were associated with the lagoon communities of October 2019 and 2020. The corresponding conditions could be characterised by a higher salinity and temperature and lower NO_3^- and O_2 concentrations. The most abundant taxa in this group were the small epipsammic taxa *S. guenter-grassii*, *N. cf. luectorii*, *O. cf. mutabilis*, *N. krumbeinii* and the epipellic taxa *Navicula gregaria*, *N. paul-schulzii* and *N. perminuta*.

The influence of changes in salinity and nutrient concentrations (PO_4^{3-} , NH_4^+ , NO_3^-) on the community compositions of the lagoon was tested in a separate CCA (Fig. 81). Salinity and nutrients together explained up to 36.0 % of the variation (inertia) within the lagoon communities. On the basis of these parameters, all three sampling dates can be separated in the CCA. The pre-flooding lagoon community was related to higher salinity and lower nutrients. In contrast, the lagoon community in January 2020 was related to elevated NO_3^- concentrations and that in October 2020 was associated with higher PO_4^{3-} and NH_4^+ levels.

The same analysis was done for the peatland communities, only in this case salinity had to be excluded due to high multicollinearity. Here the different sampling dates were well separated and the nutrient concentrations alone account for 51.4 % of the variation (inertia). Before

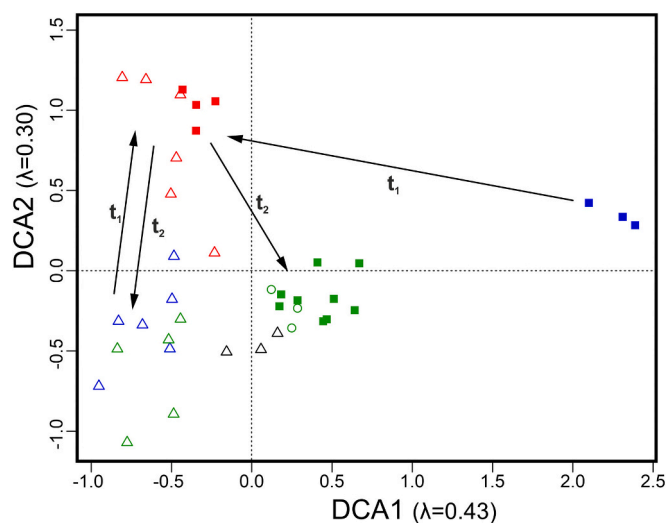


Fig. 80. Detrended correspondence analysis (DCA) of all samples. Triangles represent sampling locations in the lagoon and squares represent peatland samples. Colors indicate the sampling time: blue – October 2019, red – January 2020, green – October 2020. Time arrows indicate the changes between the sampling dates: t_1 for the change from October 2019 to January 2020 and t_2 for the change from January 2020 to October 2020. λ give the respective eigenvalues.

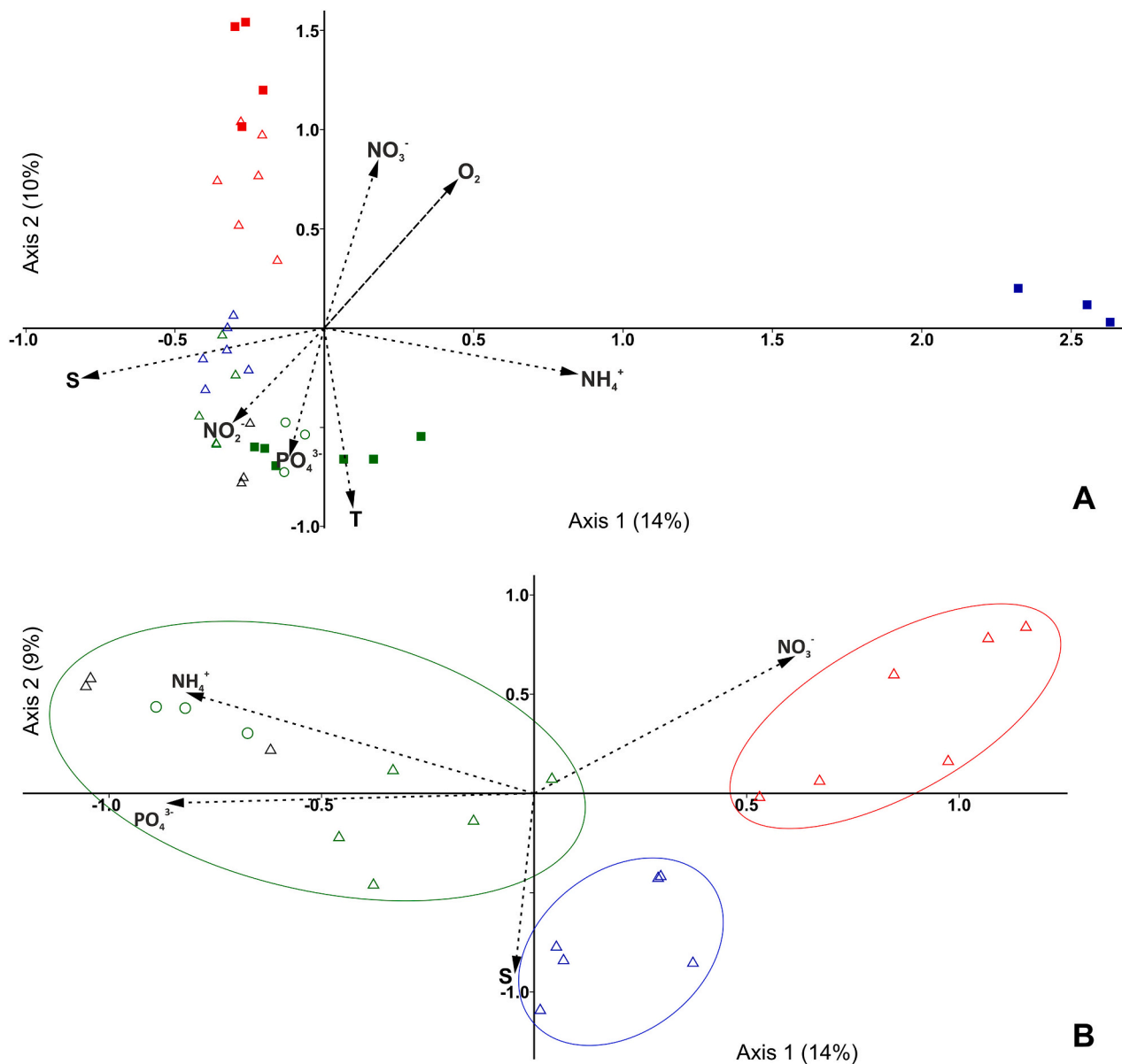


Fig. 81. Canonical correspondence analyses (CCAs) for all sites (A) and the lagoonal sites in specific (B). Triangles represent sampling locations in the lagoon and squares represent peatland samples. Colors indicate the sampling time: blue – October 2019, red – January 2020, green – October 2020. The percentages in brackets give the portion of the variation explained by the respective axis.

the flooding, the peatland community showed a relationship with high NH_4^+ concentrations. As in the case of the lagoon, the peatland community in January 2020 (here, lagoon and peatland community had the same composition) was also related to higher NO_3^- . In October 2020, the peatland community was associated with elevated PO_4^{3-} concentrations.

3.6. Benthic diatom taxa distribution and diversity

In order to evaluate which species were shared between the different phases, a Venn diagram was generated (Fig. 83). For that purpose, the data were arranged into four groups: lagoon and peatland, each before and after the flooding. As in case of the ordination analyses, it can be seen that the peatland benthic diatom community before the flooding is most different from all other groups. With 36.2 %, this group showed the highest percentage of taxa that were exclusively present within a single group. Overall, this cohort almost exclusively consisted of species which can only be found there plus ubiquitous species which were observed in all groups. This share of all groups consisted of 26 species, indicating a

portion of somewhat indifferent species. The brackish communities were more similar and, as could be expected, the largest subset contained the species shared between all brackish groups. These groups each showed a relatively high share of exclusive taxa (17.2–25.2 %), which, however, were usually found in low abundance (< 5 %). Our results illustrated that presence/absence as well as changes in abundance of taxa were important for the differences between the observed communities. The former was especially important concerning the oligohaline peatland community, while the latter might play a larger role in the case of the brackish communities. The diversity according to the calculated Gini-Simpson indices was always high (0.81–0.97). In the lagoon, no changes in this index could be observed between the different sampling times (0.95–0.97). Solely the peatland before the flooding showed a lower diversity (0.81) which increased after the flooding (0.91–0.92).

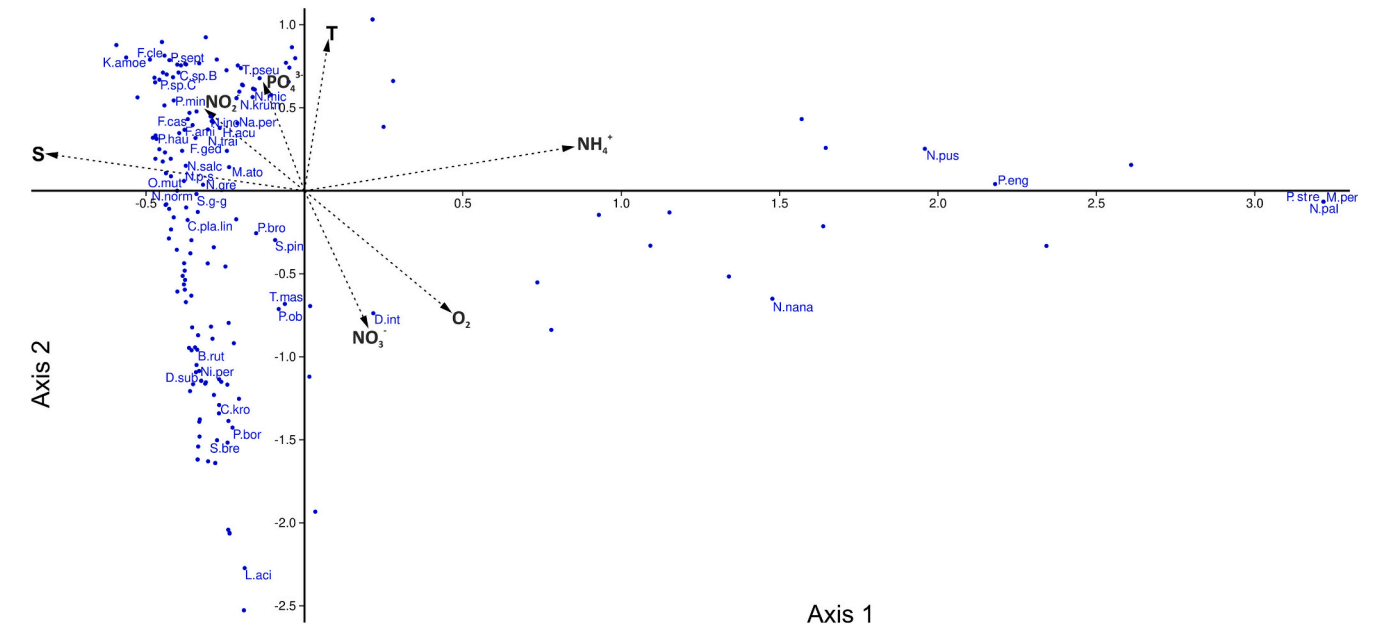


Fig. 82. Canonical correspondence analysis (CCA) for all taxa including all samples. Blue dots indicate the optima of the different taxa. Codes for taxa with a relative abundance of >5 % are given in Table 1.

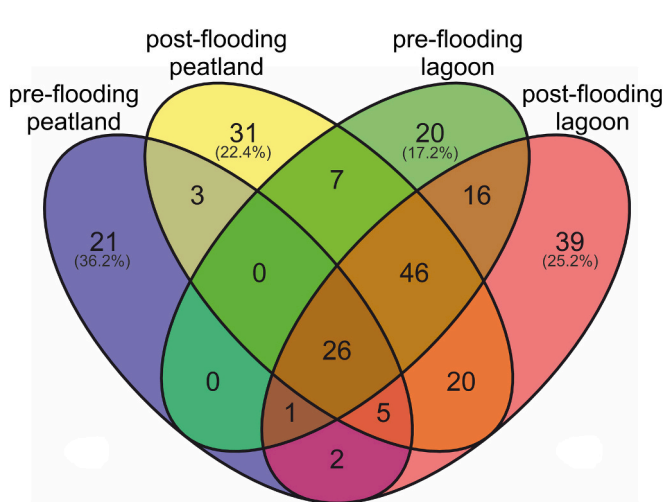


Fig. 83. Venn diagram of the peatland and lagoon diatom communities before and after the flooding. Percentages are referring to each of the four the corresponding groups.

4. Discussion

4.1. Environmental changes after rewetting

The targeted flooding of the degraded coastal peatland Drammendorf with brackish water was a sudden event with severe implications for the biogeochemistry, surface structure, hydrology and organisms of both the peatland and the lagoon. First of all, the salinity in the peatland area changed distinctly from oligohaline to mesohaline conditions, thereby providing a suitable environment for a mesohaline diatom community. Moreover, large quantities of lagoonal sediments were transported into the peatland area, facilitating, for example, the abundance of small epipsammic diatoms such as *Martyana atomus* or *Nanofrustulum krumbeinii*. Our results also further confirm the recent findings of nutrient release from the rewetted peatland by Pönisch et al. (2023) and subsequent nutrient export to the adjacent lagoon. This is especially true for dissolved inorganic nitrogen (NO_2^- , NO_3^- , NH_4^+)

immediately following the rewetting and PO_4^{3-} in October 2020. These are important nutrients for diatoms and are known to have an influence on benthic diatom community compositions. While the water column of most aquatic ecosystems shows strong seasonal fluctuations in nutrient concentrations, which are mostly depleted after spring phytoplankton blooms, sediment pore waters are typically enriched in N and P (Woelfel et al., 2009). Some benthic diatoms benefit from these high nutrient levels in the pore water, where nutrients are approximately 5–20-fold higher than in the overlying water column and are probably available year-round (Woelfel et al., 2009). The only notable change of pH concerns the peatland area, which was slightly acidic before the rewetting and was turned into a brackish, alkaline environment by the lagoonal water. Overall, the peatland underwent the most drastic changes, turning from a mostly terrestrial, oligohaline, degraded peatland with a mineralized peat layer to a widely aquatic brackish system with marine sediments. The lagoon was also distinctly influenced, but mostly by changes in its biogeochemistry. Our results also support the previous results of Pönisch et al. (2023) that rewetted coastal peatlands could be responsible for high, currently unmonitored, nutrient inputs into adjacent lagoon water, at least on a short timescale such as a few years.

4.2. Effects of rewetting on biomass

The here found range for microphytobenthic biomass of 0.9–200 mg Chl *a* m^{-2} (averages 31.2–107.0 mg m^{-2}) fits well the range of 128 ± 101 mg m^{-2} for shallow temperate coasts given by Cahoon (1999). Equally, previous results for the Southern Baltic Sea such as 28.5–87.7 mg m^{-2} by Kuriyama et al. (2021) and 64.8–138.5 mg Chl *a* m^{-2} by Karsten et al. (2021) fall into the same range. The observed, significant decrease of biomass during January 2020, shortly after the flooding, can be at least partly explained by the destruction/erosion of some part of the microphytobenthic communities due to the creation of the breakthrough followed by strong hydrodynamic forces. Both sampling points in the immediate vicinity of the breakthrough (D2, D10) exhibited very low Chl *a* concentrations and almost no diatom valves at that time. This is supported by the fact that eight of the nine samples at these locations lacked diatom valves. In the case of the remaining sample from October 2020 (D3), that lacked diatom valves, a processing error seems more likely as the other two samples from the same location contained plenty

of diatoms. In the phase of community change and repopulation following the flooding, at least in the peatland, the new mesohaline community had to build up new biomass first, which might have also contributed to a temporary decrease in Chl *a*. Notably, after nine months, in October 2020, all these sampling locations were repopulated with benthic diatoms and had a biomass comparable to the values before the flooding. This illustrates the quick generation time and response of diatom communities even to severe disturbances. Apart from the flooding, seasonality might have also played a role as the flooding coincides with the transition from autumn to winter. The average Chl *a* values for the pre- and post-flooding peatland and lagoon were higher in October 2020 than in October 2019, which might be the result of additional nutrient release from the rewetted peatland (Pönisch et al., 2023). The observed difference, however, was not statistically significant which might be due to the fact that both systems were already nutrient-rich before the flooding and that at least in October 2020 part of the phototrophic biomass in the benthos derived from a local increase of phytoplankton.

4.3. The effect of rewetting on benthic diatom communities

By assessing the diatom taxa compositions and tracking key environmental parameters it was possible to define and characterise distinct diatom communities and temporal phases. The flooding of the drained coastal peatland strongly impacted the microphytobenthos causing both short-term and long-term changes in the benthic diatom community compositions. Starting in October 2019, peatland and lagoon communities showed the highest dissimilarity, which was related to differences in salinity and nutrient concentrations. This is at least partly due to the strong spatial separation by the dyke and the more terrestrial character of the drained peatland. However, the peatland at that time was not a freshwater environment as indicated by the presence of some mesohaline taxa and a salinity just above 0.1. The nonetheless previous oligohaline peatland could be further characterised by high ammonium concentrations and some degree of desiccation over the course of the year. The latter is supported by the presence of areophilic diatom taxa. In particular, *Nitzschia pusilla* has been reported from soil samples (Stanek-Tarkowska et al., 2021; Foets et al., 2021) and *Mayamea* cf. *permitis*, a species known to be areophilic, was also abundant at this time. During an investigation of Spanish salt marshes, Trobajo et al. (2004) reported that *Nitzschia pusilla* and *N. nana* preferred higher NH_4^+ concentrations, indicating that this could be an important factor for the abundance of these taxa also in the pre-flooded peatland. Elevated ammonium levels can distinctly influence diatom community compositions (Hürlimann and Schanz, 1993; Underwood et al., 1998) and high ammonium concentrations may repress NO_3^- uptake (Glibert et al., 2016, and references therein). On the other hand, high ammonium concentrations in combination with high light and temperature conditions and pH above 7 will lead to enhanced NH_3 levels which might become toxic for different microalgae (Natarajan, 1970; Hürlimann and Schanz, 1993; Andersen et al., 2019). The pH in the peatland was below 7 in October 2019, but increased to 7.7–8.0 after rewetting, and hence the risk of NH_3 formation might increase particularly during summer when light and temperature levels are enhanced.

In January 2020, an immediate impact of the flooding on the diatom community compositions could clearly be detected. Apart from physical forces eroding the microphytobenthos from the sediments/soils in the vicinity of the breakthrough, both lagoon and peatland communities changed distinctly to become the same brackish community. This change was most drastic for the peatland benthic diatoms which were completely transformed from an oligohaline environment to a permanently flooded brackish environment. Likewise, the community composition changed completely, and strong gradients in salinity and NH_4^+ concentrations are possible drivers for these conspicuous effects. However, the lagoonal community also distinctly changed. As in previous studies, changes in nutrient concentrations and salinity were

found to be important factors in the CCA (e.g. Snoeijis, 1994; Ulanova and Snoeijis, 2006; Ulanova et al., 2009). However, the fact that the flooding coincided with the seasonal change from autumn to winter complicates the interpretation concerning chlorophyll *a* and nutrient concentrations. In case of the chlorophyll *a* content, and as already discussed above, during the dyke opening the flooding event had a negative impact on photosynthetic biomass but also during the follow-up phase of community changes. Nevertheless, some species may benefit from the hydrodynamic disturbance and changing environmental conditions as was suggested particularly for small opportunistic fragilarioid taxa (Sullivan and Moncreiff, 1988; Weckström and Juggins, 2006). Those are certainly abundant after the flooding of the peatland, but generally occur with high cell numbers under brackish conditions. In addition, according to the CCA, only biraphid taxa showed an optimum at this time, more precisely the aerophilic taxa *Pinnularia borealis* agg., *Chamaepinnularia krookiformis* and *Luticola acidoclinata* as well as *Berkeleya rutilans*, *Nitzschia* aff. *perindisticta*, *Denticula subtilis* and *Surirella brebissonii*. At least the aerophilic taxa might be more tolerant towards physical disturbance and fluctuating water levels compared to aquatic species, but it has to be noted that the contribution of seasonality cannot be clearly separated. The same is true for the elevated NO_3^- concentrations, which, to some extent, would be expected during winter due to nitrification (Voss et al., 2010). At the same time, nutrients from the degraded peat are also released (Pönisch et al., 2023).

In October, both communities changed again and photosynthetic biomass reached the pre-flooding level. And while the peatland community was again distinct, the lagoon community exhibited a composition similar to that before the flooding, indicating that the effect of the flooding on the microphytobenthos of the lagoon was relatively short-lived. Solely the area immediately in front of the dyke breakthrough was still distinctly influenced by the peatland and showed an intermediate species composition. The local abundance of planktonic diatom species within the peatland area in October 2020 may be caused by the PO_4^{3-} released from the peat itself as these taxa revealed a relationship with increased PO_4^{3-} levels in the CCA. Weckström and Juggins (2006) also found a relation between the occurrence of the planktonic taxa *Thalassiosira pseudonana* and *Cyclotella atomus* and total phosphorous level. The abundance of planktonic diatoms did not lead to a decrease in the microphytobenthic biomass as the benthic chlorophyll *a* concentrations in the peatland at that time were on average the highest of all measurements (48–163 mg m^{-2} , Fig. 2).

Salinity effects on benthic diatom communities are best documented over larger salinity gradients (Weckström and Juggins, 2006), which is also confirmed here when comparing the pre- and post-flooding peatland. However, the interactions, especially in a highly dynamic event such as the studied flooding, are most complex and the role of other factors remains unclear. Factors that might have an additional role in this case include grazing and other biotic interactions, water table, turbulence/exposure, light intensity, desiccation, substrate, organic carbon content and silica concentrations. Nevertheless, the variation explained by the measured parameters is quite high (40.9 %) and in the same range as in previous studies on coastal diatoms (Zong and Horton, 1999; Weckström and Juggins, 2006; Sawai et al., 2016).

Overall, the community changes in the lagoon were transitory while the change of the peatland community was permanent. Despite being closely connected to the rest of the lagoon, the diatom community of the peatland in October 2020 remained different from the lagoon community. This indicates that coastal peatlands have unique diatom communities, possibly also under more natural conditions (neither degraded nor completely flooded). It seems reasonable to assume that peat is a key factor for this difference as peatland derived organic compounds can stimulate growth in diatoms (Prelle and Karsten, 2022). In addition, the CCA points to species which could be specialists for this unique combination of peat and brackish water, namely *Nitzschia nana*, *N. cf. pusilla*, *Planothidium engelbrechtii*, *Pinnularia streckeriae* and *Fragilaria famelica*. The fact that the nutrient release from the peatland did not permanently

alter the lagoon community could be explained by the already highly eutrophic status of this water body.

4.4. Diversity and taxonomy of the benthic diatom communities

Despite the small spatial scale of this study, the total number of 245 taxa can be considered as relatively high. It exceeds the numbers of some previous studies on coastal environments of the Baltic Sea on similar to larger spatial scales (Vilbaste et al., 2000; Virta and Soininen, 2017; Witak et al., 2020), which reported between 133 and 230 taxa. Other more large-scale studies, as could be expected, described more taxa (Busse and Snoeijs, 2002; Båk et al., 2004; Ulanova and Snoeijs, 2006; Ulanova et al., 2009; Virta et al., 2020; Virta, 2023). The diversity according to the Gini-Simpson indices was high and only slightly lower in the pre-flooding peatland. This index may be interpreted as the probability that two randomly picked valves in sampling area belong to a different species. Apart from that, it has to be considered that diatoms are microorganisms and even taxa with a very low count in our samples may occur in very high total numbers in the habitat. Moreover, the lack of sampling of different microhabitats may also impair diversity measurements in this case. Our data also highlight a high biodiversity of benthic diatoms under brackish conditions which are often considered as less diverse due to the so-called Remane diagram (Remane, 1940), which shows how aquatic species diversity changes from freshwater to more marine areas. Also previous studies could not confirm a reduced benthic diatom biodiversity under brackish conditions (Snoeijs, 1994). To the best of our knowledge the presented results are also the first study on recent diatom communities of a Baltic Sea coastal peatland, despite the importance and abundance of this biotope type along the southern Baltic Sea coast. Concerning the lagoon benthic diatom communities, most of the abundant taxa were already reported in previous studies of the southern Baltic Sea coast (e.g. Witkowski et al., 2000; Witak, 2013; Witak et al., 2020). Nevertheless, there are some taxa which were not previously reported and taxa for which the identification remains uncertain.

Among the species with an abundance >5 %, five taxa could not be identified on species-level (*Karayevia* aff. *amoena*, Figs. S73–S82; *Nitzschia* sp. “A”, Figs. S49–S55; *Nitzschia* aff. *microcephala*, Figs. S25–S43; *Cocconeis* sp. “B”, Figs. S83–S88; *Planothidium* sp. “C”). Considering that the taxonomic literature on coastal and brackish diatoms is rather limited compared to, for example, benthic freshwater diatoms, identification is challenging. In some cases, the observed morphological variability exceeds the descriptions in the literature. For *Fragilaria casubica*, for example, some valves fit the description by Witkowski et al. (2000) while others are too small or have a lower stria density. The description of *Fragilaria gedanensis* (Witkowski et al., 2000) is sparse and the population in the present study exhibited a higher stria density (16–18 instead of 9–14 in 10 µm, Figs. 26–30). *Chamaepinnularia krookiformis* was not differentiated from *C. plinskii* Zelazna-Wieczorek & Olszynski in this study as both morphotypes always occurred together and intermediate forms were also present (Figs. 70–75).

Concerning *Nanofrustulum* cf. *luectorii* (Figs. 56–72), the light microscopical characters fit the description of *N. luectorii* Morales, Novais, Garcia, Maidana & Morais, a so far poorly known taxon which was described as epilithic, possibly aerophilic from shallow, temporary rivers in southern Portugal (Morales et al., 2023). However, ultrastructure features may still reveal differences. The ecology of this taxon remains widely unknown. Likewise, in the case of some small araphid taxa such as *Nanofrustulum trainorii* (Figs. 16–20) or *Opephora* cf. *mutabilis* (Figs. 21–25), some residual uncertainty remains concerning the morphological identification without the use of scanning electron microscopy. However, the issue of taxonomic resolution extends beyond morphology as molecular approaches revealed hidden or even cryptic species level diversity within many diatom groups (e.g. Jahn et al., 2017; Pinseel et al., 2020; De Luca et al., 2021; Schultz et al., 2022, 2024) and therefore a purely morphological assessment will most probably

underestimate species diversity. A well-known example from our data set is the *Pinnularia borealis* group which was found to encompass at least 126 different lineages based on molecular data (Pinseel et al., 2020). There are also suspicions that *Mayamaea* cf. *permitis* may in fact be a species complex or cryptic species (Bíró et al., 2022). Therefore, an integrated morphological and molecular approach is required for better understanding the dynamics and diversity of coastal benthic diatom communities. On the other hand, the resulting taxonomic progress creates problems concerning the comparability of floristic and ecological studies as well as data on species ecology over time (e.g. Van Dam et al., 1994; Vos and de Wolf, 1993). Consequently, constant taxonomic revisions as well as better knowledge on the diversity and taxonomy of coastal benthic diatoms are required, particularly when using the data for bioindication of coastal flooding events.

5. Conclusions

This study provides a comprehensive species inventory for benthic diatoms of a coastal peatland and adjacent lagoon of the southern Baltic Sea, thereby contributing to the knowledge of biodiversity of these ecologically important microalgae. Particularly for coastal peatlands huge knowledge gaps exist on diversity and taxonomy of benthic diatom communities. Due to their short generation times and their ecological and taxonomic diversity, diatoms are well suited and widely used as indicators of environmental conditions and changes. Diatom community composition data are well suited to track changes in dynamic coastal environments such as changes in salinity, trophic status, substrate and sea level (Ryves et al., 2004; Horton et al., 2006; Weckström and Juggins, 2006; Taffs et al., 2008).

This paper documented that the diatom communities in the study area distinctly responded to environmental changes associated with the sudden flooding event. Therefore, these benthic diatoms may be used as bioindicators to track coastal flooding events and monitor the ecological state of coastal peatlands along the southern Baltic Sea coast. For an effective use of diatoms as bioindicators of dynamic coastal processes, further research on the diversity, taxonomy and ecology of coastal diatoms species is needed.

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CRediT authorship contribution statement

Konrad Schultz: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Mirko Dreßler:** Writing – review & editing, Visualization, Formal analysis. **Ulf Karsten:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Petra Thea Mutinova:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Lara R. Prella:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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